Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 6 N force acts at 120° to a 5 m displacement. Calculate its work and explain the sign.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and record actual trials separately.

Worked reasoning: Supplied measurements give net work 12 J and kinetic-energy change 11 J. Their 1 J difference could reflect uncertainty or omitted work; it is evidence to investigate.

Your turn: Draw two bars for these measurements and label one check of the experiment.

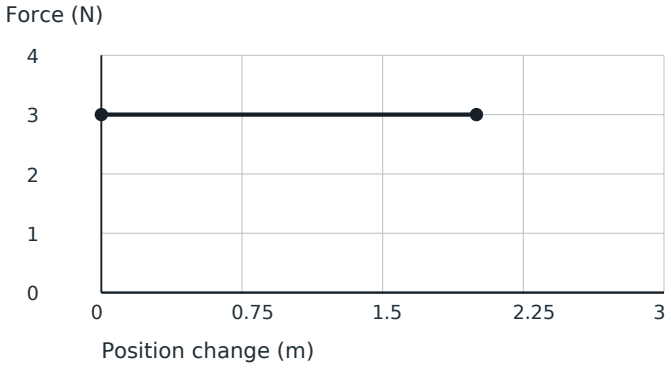
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

The supplied force-displacement graph stays at 3 N over 2 m. Calculate work.



Marzano check - circle: ① ② ③

04 Apply the physics

A 0.50 kg lab cart speeds up from 0.40 to 1.20 m/s. Measured pulling work is +0.35 J. Complete the energy account. Find ΔK and infer the work of resistance if every other force does zero work.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Create a trial table with mass, initial/final speed, work and ΔK. Name two causes of disagreement that deserve checking.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 0.4 kg projectile at 60 m/s sticks to a stationary 3.6 kg cart. Find common speed and the fraction of initial kinetic energy remaining.
Use before and after snapshots to organize the momentum account.

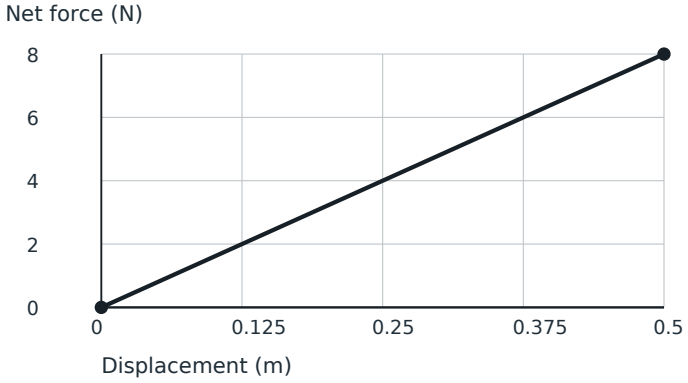
Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

07 Exit check

A force rises linearly from 0 to 8 N over 0.50 m. Find work. If a 1 kg cart begins at rest and this is net work, find final speed. Use the energy model to show the quantities you compare; label units and any unknowns.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

A 2 kg cart speeds from 1 to 3 m/s while measured pulling work is 10 J. Infer friction work and mean friction magnitude over 2 m if other work is zero. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③